

INFORMATION TECHNOLOGY 110 LAB E

LAB ASSIGNMENT 2

Due: Jan 30, 11:59 pm.

Topics:

Signed and Unsigned Bits

One's Complement

Two's Complement

Binary Addition

1. Given a register of $n=8$ bits,
 - a. How many integers can be represented in this register?
 - b. What is the range of unsigned integers that can be represented?
 - c. What is the range of one's complement signed integers that can be represented in this register?
2. Assume there is a class of 20 students and the students ID range is from 1 to 20.
 - a. What is the minimum number of bits needed so that each student's ID is assigned a unique bit representation in a register?
 - b. If more students want to take the class and their ID numbers range from 21 to 40, can these new ID numbers be represented in bits using the same bit size in 2a.? Explain the answer.
3. Perform the following operations on unsigned binary numbers and show the work:
 - a. $1110 + 101$
 - b. $11110 * 110$
4. Show the 16-bit 2's complementary binary representation for the decimal numbers 1500 and -1500.
5. Add the following two 12-bit binary two's complement numbers. Show the result in binary and decimal. Note: The binary result may contain a carry-over.

$$\begin{array}{r} 0110\ 0101\ 0111 \\ + \underline{1110\ 0110\ 0110} \end{array}$$

Submit the answers for grading to Moodle.